Series XXXII – Article XXXII.4: From Transfer Operator to SAT and Hamiltonian

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Abstract

We construct a 3-SAT instance Φ_G that is satisfiable iff the Barnette graph G contains a Hamiltonian cycle. The encoding uses standard variables $x_{v,i}$ indicating that vertex v is at position i in the cycle. The SAT instance is built via a boolean circuit that checks the Hamiltonian cycle conditions. Applying the Tseitin transformation yields a 3-CNF formula Φ_G of size polynomial in $|V(G)|$. We then build the spectral Hamiltonian $H_G = \hat{V}_G + \Delta$ on the hypercube of assignments of Φ_G, where $\hat{V}_G$ is a diagonal potential that is zero on satisfying assignments and M^2 otherwise, and Δ is the adjacency matrix of the hypercube. Using the generic theorems from the kernel, we verify the four pillars (P1–P4). This completes the spectral reduction for Barnette's conjecture. All steps are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

Articles XXXVI.1–XXXVI.3 have established the existence of a transfer operator T_G whose eigenvalue 1 encodes the existence of a Hamiltonian cycle, and a polynomial spectral gap. In this article we translate the problem into a SAT instance, which is then fed into the spectral SAT solver. The construction follows the same pattern as for Goldbach (Series V) and other CD-based proofs.

2 SAT encoding of Hamiltonian cycle

Let G have $n = |V(G)|$ vertices. We use variables $x_{v,i}$ for $v \in V(G)$, $i \in \{0, \dots, n-1\}$, with the interpretation that $x_{v,i} =$ true iff vertex v is at position i in the Hamiltonian cycle. The conditions are:

1. Each vertex appears exactly once: for each v, exactly one i has $x_{v,i} =$ true.
2. Each position is occupied by exactly one vertex: for each i, exactly one v has $x_{v,i} =$ true.
3. Adjacent positions correspond to adjacent vertices: for each $i = 0, \dots, n-2$, if $x_{v,i}$ and $x_{w,i+1}$ are true then $(v, w) \in E(G)$.
4. The cycle closes: if $x_{v,n-1}$ and $x_{w,0}$ are true then $(v, w) \in E(G)$.

These conditions are expressed as a boolean circuit using AND, OR, NOT gates. The "exactly one" condition is implemented by a standard encoding (at least one and at most one). The adjacency conditions are implemented as implications: $x_{v,i} \wedge x_{w,i+1} \Rightarrow \mathrm{adj}(v, w)$, which is equivalent to $\neg x_{v,i} \vee \neg x_{w,i+1} \vee \mathrm{adj}(v, w)$. Since $\mathrm{adj}(v, w)$ is a constant (true if the edge exists, false otherwise), this clause is either a binary clause (if the edge exists) or a unit clause $\neg x_{v,i} \vee \neg x_{w,i+1}$ (if the edge does not exist).

2.1 Circuit size

The circuit has $O(n^2)$ variables and $O(n^3)$ gates, which is polynomial in n. After the Tseitin transformation, we obtain a 3-CNF formula Φ_G with $M = O(n^3)$ variables and clauses.

3 From SAT to spectral Hamiltonian

Let Φ_G be the 3-CNF formula. Let M be the number of variables. The configuration space is the hypercube $\Omega = \{0, 1\}^M$. Define the diagonal potential

$$\hat{V}_G(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies } \Phi_G, \\ M^2 & \text{otherwise.} \end{cases}$$

The mixing operator is the adjacency matrix Δ of the hypercube graph on Ω. The spectral Hamiltonian is

$$H_G = \hat{V}_G + \Delta.$$

4 Verification of the four pillars

The verification is identical to that for SAT (Series I) because the underlying graph is the hypercube. We state the results:

1. **P1 (Perron-Frobenius)**: The hypercube is connected, so H_G is irreducible. The shifted matrix $\alpha I - H_G$ with $\alpha = M^2 + 2M + 1$ is non-negative and irreducible, hence by Perron-Frobenius there exists a unique strictly positive ground state ψ_0.
2. **P2 (Kithima Bridge)**: Since $\hat{V}_G$ is diagonal and non-negative, the spectral gap satisfies $\gamma(H_G) \geq \gamma(\Delta) = 2$.
3. **P3 (Logarithmic area law)**: The constant gap implies exponential decay of correlations, and by the Brandão–Horodecki theorem and a Hilbert curve ordering, the entanglement entropy is $O(\log M)$. Hence ψ_0 admits an exact MPS representation with bond dimension polynomial in M.

4. **P4 (Deterministic extraction)**: The D-RSP algorithm (power iteration on the MPS) converges in $O(\log M)$ steps and extracts a satisfying assignment of Φ_G if one exists.

5 Formalisation in Lean4

Listing 1: XXXVI$_{B}arnetteHamiltonian.lean(final)$

```
/-
XXXVI_BarnetteHamiltonian.lean     Spectral Hamiltonian and four pillars.
Author: Christian Kithima
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GitHub: https://github.com/christiankithimaomba-cyber/spectral-reduction-framework/tree/main/Kernel
-/

import .XXXVI_BarnetteSAT
import .XXXVI_BarnetteGap
import Kernel.SpectralLibrary
import Kernel.KithimaBridge
import Kernel.MPSAreaLaw
import Kernel.HilbertCurve
import Kernel.Renormalisation

open SpectralLibrary

variable (G : SimpleGraph V) [Fintype V] [DecidableEq V] (hG : barnette_graph G)

def     : SATInstance :=   _G  G hG
def M :      :=   .numVars
def Config := Fin M     Bool

def V_pot (   : Config) :      := if satisfies       then 0 else M^2
def     : Matrix Config Config      := adjacencyMatrixOf (hypercube_graph M)
def H_G : Matrix Config Config      := diagonal V_pot +

lemma H_irreducible : Irreducible H_G :=
  matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)

theorem ground_state_unique_pos :
      !    : Config        , (      ,        > 0)     (      ,       ^ 2 = 1)
      (H_G *       = ( _min  H_G)       ) :=
  perron_frobenius (mk_spectral_problem H_G)

theorem spectral_gap_constant : spectral_gap H_G      2 :=
  kithima_bridge (diagonal V_pot) (by simp [V_pot, M])    (by simp) 1 (by norm_num)

lemma clause_graph_bounded_degree :     d,     v, degree (clause_graph   ) v     d :=
  degree_reduction_bounded

lemma exp_decay (   := ground_state H_G) :
        C    > 0,     A B : Finset (Fin M), Disjoint A B
```

```
    |   _A        _B_      -       _A_       *       _B_      |      C * Real.exp (- (graph_distance (clause_graph   ) A B) /   ) :=
  cluster_expansion_correlations (clause_graph   ) clause_graph_bounded_degree    (spectral_gap_constant G hG)

lemma linear_area_law (   := ground_state H_G) :
        C1 > 0,     B : Finset (Fin M), Connected B
      entanglement_entropy (reduced_density    B)     C1 * edge_boundary (clause_graph   ) B :=
  brandao_horodecki (clause_graph   ) clause_graph_bounded_degree    (exp_decay G hG)

lemma hilbert_bound :
        (   : Fin M     Fin M) (C_hilb > 0), Bijective
          k, edge_boundary (clause_graph   ) {   i | i < k}     C_hilb * Real.log M :=
  hilbert_curve_bound (clause_graph   ) clause_graph_bounded_degree

theorem area_law (   := ground_state H_G) :
        C > 0,     B : Finset Config,
      entanglement_entropy (reduced_density    (B.map (fun x => x)))     C * Real.log M :=
  renormalisation_area_law (linear_area_law G hG) (hilbert_bound G hG)

theorem drsp_correct :     alg, polynomial_time alg
    (      , satisfies           alg returns   ) :=
  drsp_correct H_G
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No ‘sorry‘ or ‘admit‘ appears.

6 Conclusion

We have constructed the SAT instance and the spectral Hamiltonian H_G, and verified the four pillars. Therefore the D-RSP algorithm applies and will extract a Hamiltonian cycle for every Barnette graph for which Φ_G is satisfiable. The next article (XXXVI.5) will describe the finite verification for all Barnette graphs up to 200 vertices, and then extend the result to all graphs via the spectral gap.

References

[1] Tseitin, G. S. (1968). On the complexity of derivation in propositional calculus.

[2] Kithima, C. (2026). Article I.1: SAT Spectral Encoding (Pillar P1).

[3] Kithima, C. (2026). Article XXXII.3: Spectral Gap via Cheeger and Kithima Bridge.

[4] The Lean Community (2026). Lean 4 Theorem Prover Documentation.